

The University of Texas at Austin
IN-CLASS WORK 3

M339G Predictive Analytics

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F — DISTRIBUTION.

MOTIVATION.

Assume a normal population model with both parameters unknown.

PROBLEM 3.1: *What is the unbiased estimator of the population variance?*

SOLUTION:

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

PROBLEM 3.2: *Find a pivotal quantity for σ^2 .*

SOLUTION:

$$(n-1) \frac{S^2}{\sigma^2} = \sum_{i=1}^n \left(\frac{X_i - \bar{X}}{\sigma} \right)^2 \sim \chi^2(df = n-1)$$

PROBLEM 3.3: *When dealing with two independent normal samples with unknown parameters, in broad strokes, how would you test*

$$H_0 : \sigma_1 = \sigma_2 \text{ against } H_a : \sigma_1 \neq \sigma_2,$$

i.e., which test statistic would you choose?

SOLUTION: *Note that*

$$\sigma_1 = \sigma_2 \Leftrightarrow \frac{\sigma_1}{\sigma_2} = 1.$$

It would be convenient to consider the test statistic $\frac{S_1^2}{S_2^2}$.

DEFINITION.

DEFINITION 3.4: Let U and V be independent, chi-squared random variables with degrees of freedom ν_1 and ν_2 , respectively. Then, the random variable

$$F = \frac{U/\nu_1}{V/\nu_2}$$

is said to be **F-distributed** with ν_1 numerator and ν_2 denominator degrees of freedom. We write

$$F \sim F(\nu_1, \nu_2) = F_{\nu_1, \nu_2} .$$

APPLICATION.

THEOREM 3.5: Let two independent random samples of sizes n_1 and n_2 be drawn from two normal populations with variances σ_1^2 and σ_2^2 , respectively. Denoting their respective sample variances by S_1^2 and S_2^2 , we have

$$F = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F_{n_1-1, n_2-1} .$$

COROLLARY 3.6: If $\sigma_1 = \sigma_2$, then

$$F = \frac{S_1^2}{S_2^2} \sim F_{n_1-1, n_2-1} .$$

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